

Chapter 7: Teaching Notes

Three 55-minute classes

Chapter 7 — Class 1

Externalities: private decisions and social costs

Instructor preparation

Length: 55 minutes. **Slides:** 1–15. **Reading:** Chapter 7 notes, “An Introduction” through “Exercise — Identifying and Correcting Externalities.”

Learning goals: students can identify a third party, distinguish private from social marginal values, calculate private and efficient quantities, and explain how a corrective tax or subsidy changes incentives.

Board setup: reserve the left side for definitions, the middle for the negative-externality graph, and the right side for calculations. Erase the middle and right for the positive-externality example. Keep these two rules visible:

Private market: $MPB = MPC$ Efficient quantity: $MSB = MSC$.

Say: “Marginal means the effect of one more unit.” Define each abbreviation before using it. The examples assume competitive markets and constant spillovers per unit. Government administration and enforcement costs are omitted.

0–5 minutes | Opening: who is missing? — slides 1–3

Say: “A buyer and a seller may both gain from a trade. A third person may still be harmed. We need to count that person’s cost.”

Use a refinery selling fuel. Ask: “Who buys? Who sells? Who may experience a cost without taking part in the sale?”

Expected answer: fuel buyer, refinery, nearby residents affected by pollution. Take two responses. Separate residents’ pollution damage from the buyer’s spending on fuel.

Then show slide 3. Ask for a quick vote on whether zero pollution is always the best economic target. Do not settle the question yet.

Bridge: “We will compare the benefit and cost of one more unit. This gives us a way to choose a quantity.”

5–12 minutes | The language of externalities — slides 4–9

Write and say:

- Private cost: a cost considered by the participant.
- External cost: a cost imposed on another person and omitted from the participant's incentives.
- Social cost: private cost plus external cost.

$$MSC = MPC + MEC, \quad MSB = MPB + MEB.$$

Read the first equation aloud: “Marginal social cost equals marginal private cost plus marginal external cost.” Do the same for benefits.

Say: “Market failure means the market does not maximize total net benefits. A firm can earn a profit in a market that is inefficient.”

Give students 30 seconds to identify the third party in each example:

Activity	Private effect	External effect
Driving at rush hour	Driver's own time and fuel	Extra delay imposed on other drivers
Education	Student's skills and earnings	Knowledge that benefits other people

Ask: “Is a student's higher salary itself an external benefit?” **Answer:** no; it is a benefit received by the participant.

Use slides 8 and 9 as quick checks. Expected answers: smoking when others experience smoke; college education when it generates benefits for others. Require a named third party. The activity label alone is not the explanation.

12–27 minutes | Board example 1: the refinery — slide 10

Allow about 3 minutes for setup and the private equilibrium, 5 for the social optimum, 4 for the graph, and 3 for interpretation and responses.

Write: “Illustrative market; quantity in units; prices and marginal values in dollars per unit.”

$$D : P = 14 - Q, \quad S : P = 2 + Q, \quad MEC = 4.$$

There is no external consumption benefit, so demand is both private and social marginal benefit.

Step 1: find the private market outcome

Ask: “Which two curves guide buyers and sellers?” Then solve:

$$\begin{aligned} 14 - Q &= 2 + Q \Rightarrow 12 = 2Q \Rightarrow Q_M = 6, \\ P_M &= 14 - 6 = 8. \end{aligned}$$

Say: “At six units, the value to the buyer equals the private cost to the seller. The market calculation leaves out pollution damage.”

Step 2: count the external cost

$$MSC = (2 + Q) + 4 = 6 + Q.$$

Ask students to predict whether the efficient quantity will be above or below six before solving:

$$14 - Q = 6 + Q \Rightarrow 8 = 2Q \Rightarrow Q_S = 4.$$

Say: “Including the damage raises the cost of each unit. Four units maximize total net benefits in this example.”

Step 3: draw the graph

1. Label the horizontal axis “Quantity” and the vertical axis “Dollars per unit.”
2. Draw downward demand using points (0, 14) and (6, 8); label it $D = MPB = MSB$.
3. Draw upward private supply using (0, 2) and (6, 8); label it $S = MPC$.
4. Draw parallel social marginal cost using (0, 6) and (4, 10); label it MSC .
5. Mark private equilibrium (6, 8) and the social intersection (4, 10). Drop lines to quantities 6 and 4.
6. Draw a vertical bracket of 4 between the two cost curves at the same quantity. Label it “external cost per unit.”

Do not call 10 the unregulated market price. It is the marginal social value and cost at the efficient quantity. It becomes the buyer price under the tax introduced next.

Step 4: explain the excess output

Ask: “At $Q = 5$, what is the benefit? What is the full cost?”

$$MSB(5) = 9, \quad MPC(5) = 7, \quad MSC(5) = 11.$$

Say: “The buyer values this unit at nine dollars. The seller’s private marginal cost is seven. But the full marginal cost is eleven. Private incentives support output that costs society more than it is worth.”

“Too much production” means more than the efficient quantity. All six units may sell; this is not an unsold surplus.

27–37 minutes | Corrective tax and welfare — slides 11–13

Ask: “How could we make the seller face the omitted four-dollar cost?”

Write a tax of $t = 4$ per unit. In terms of the buyer price, taxed supply is:

$$P_B = 2 + Q + 4 = 6 + Q.$$

Demand meets taxed supply at $Q = 4$. Then calculate both prices:

$$\begin{aligned} P_B &= 14 - 4 = 10, & P_S &= 2 + 4 = 6, \\ P_B - P_S &= 4, & \text{tax revenue} &= 4 \times 4 = 16. \end{aligned}$$

Say: “Buyers pay ten. Sellers keep six after the tax. The four-dollar difference is paid to government.”

Ask: “Does charging the seller mean only the seller bears the burden?” **Answer:** no. Relative to the original price of 8, buyers pay 2 more and sellers receive 2 less in this example.

Shade the triangle between demand and MSC from $Q = 4$ to $Q = 6$. Its vertices are $(4, 10)$, $(6, 8)$, and $(6, 12)$.

$$DWL = \frac{1}{2}(6 - 4)(12 - 8) = 4.$$

Say: “This area is the net loss from excess output. The tax removes that loss in our model. The sixteen dollars of revenue is a transfer, not the welfare gain.”

Remaining damage is $4 \times 4 = 16$. Efficient output does not require zero damage here. The resource benefits from the first four units justify their full costs.

Use slide 13: subsidizing production of the harmful good generally moves output in the wrong direction in this model. Regulation can also reduce output, but its design and costs matter.

Quick check: “Does simply knowing about the damage guarantee internalization?” **Answer:** no. Incentives must account for the damage.

37–49 minutes | Board example 2: a positive spillover — slides 14–15

Say: “Now suppose an activity benefits other people. The buyer does not count the whole benefit. The market may provide too little.”

Use education or another service with an assumed external benefit. Keep the numbers from the notes:

$$MPB = 10 - Q, \quad MPC = MSC = 2 + Q, \quad MEB = 4.$$

Give students one minute to calculate the private quantity, then work through the result:

$$10 - Q = 2 + Q \Rightarrow Q_M = 4, \quad P_M = 6.$$

Include the benefit to others:

$$MSB = 14 - Q, \quad 14 - Q = 2 + Q \Rightarrow Q_S = 6.$$

Draw: downward MPB with intercept 10; downward MSB with intercept 14; upward MSC with intercept 2. Mark private intersection $(4, 6)$ and social intersection $(6, 8)$. Mark the vertical benefit gap of 4. Ask students which quantity is larger and why.

At $Q = 5$, private benefit is 5, social benefit is 9, and cost is 7. Say: “This additional output creates a social gain. Private willingness to pay is too low to support it without a change in incentives.”

A subsidy of $s = 4$ creates a difference between the seller’s receipt and the buyer’s payment:

$$P_S = 2 + 6 = 8, \quad P_B = 10 - 6 = 4, \\ P_S - P_B = 4, \quad \text{government spending} = 4 \times 6 = 24.$$

Say: “The seller receives eight dollars per unit. The buyer pays four. Government pays the other four. The external benefit existed before the subsidy.”

If students are keeping pace, shade the missed-gain triangle with vertices $(4, 6)$, $(4, 10)$, and $(6, 8)$; its area is 4. Otherwise state the direction of the welfare effect and keep the exit check on time. Government spending of 24 is not a social-surplus gain of 24.

49–55 minutes | Exit check and transition

Allow three minutes of individual work and three minutes to review.

Prompt: Demand is $18 - Q$, private supply is $2 + Q$, and marginal external cost is 4.

1. Find the private quantity and efficient quantity.
2. Find the corrective tax and the two prices after the tax.
3. Finish: “The private quantity is too high because ...”

Answer key: $Q_M = 8$, $P_M = 10$; $MSC = 6 + Q$; $Q_S = 6$; tax 4; buyers pay 12, sellers receive 8. The market omits costs imposed on third parties. Optional check: initial DWL is $\frac{1}{2}(8 - 6)4 = 4$.

Closing sentence: “Next time we will ask when people can bargain over these effects and how access to a good changes incentives.”

Pacing and optional depth

If behind: omit tax incidence and the subsidy triangle. Keep both quantity calculations, the tax price wedge, and the exit check. If algebra is slow, complete only the negative-externality calculation and explain the positive case with the graph.

Optional extension, outside the plan: verify the tax’s four-dollar welfare gain by accounting for all components. Before tax: consumer surplus 18, producer surplus 18, damage 24, net surplus 12. After tax: consumer surplus 8, producer surplus 8, revenue 16, damage 16, net surplus 16. Revenue is counted once; damage is subtracted once.

Misconceptions to listen for: social means government; all personal benefits are external; over-production means unsold goods; tax revenue is deadweight loss; social efficiency means no remaining pollution.

Chapter 7 — Class 2

Property rights, bargaining, and types of goods

Instructor preparation

Length: 55 minutes. **Slides:** 16–28. **Reading:** Chapter 7 notes, “Property Rights and Incentives” through “Exercise — Rights and Types of Goods.”

Learning goals: students can explain how ownership changes incentives, identify an efficient bargain under alternative rights, distinguish efficiency from distribution, and classify goods using rivalry and excludability.

Board setup: left side for a cost comparison and bargaining payments; right side for the two-by-two goods matrix. Keep “Who can be excluded?” and “Does one person’s use reduce availability for others?” visible.

0–5 minutes | Retrieve last class

Ask students to answer without notes:

1. “Why can private equilibrium differ from the efficient quantity?”
2. “For a negative externality, which cost is missing?”
3. “Must government always be the party that changes incentives?”

Answers: third-party effects may be omitted; marginal external cost; not always—affected parties may negotiate when rights are clear and bargaining is feasible.

Say: “Today we ask who controls a resource and who can benefit from it. These rules affect decisions.”

5–11 minutes | Ownership and incentives — slides 16–17

Use a privately owned bicycle. Write four verbs: **maintain, protect, conserve, trade.**

Ask for one action under each verb. Expected examples: repair brakes, lock the bicycle, avoid unnecessary wear, sell it to someone who values it more.

Say: “An owner receives many benefits of care and bears many costs of neglect. That creates an incentive to care for the resource.”

Then ask: “If I own my car, do I automatically consider all the delay I impose on other drivers?”

Answer: no. Ownership can improve care while leaving external effects unresolved.

Define **property rights** as rules specifying who can control or use a resource. Rights need to be clear and enforceable to guide a bargain.

11–27 minutes | Board example 1: cattle and crops — slides 18–20

Use about 3 minutes for the resource comparison, 8 for the two rights cases, and 5 for the expensive-fence variation.

Write:

Annual crop damage without fence = \$1,000

Annual fence cost = \$600

Assume the fence prevents all damage, cattle production continues, no cheaper remedy exists, and bargaining has no cost. All figures are annual so they can be compared directly.

Step 1: find the efficient physical outcome

Outcome	Fence cost	Crop damage	Total resource cost
No fence	0	1,000	1,000
Fence	600	0	600

Ask: “Which outcome uses fewer resources?” **Answer:** build the fence; net gain is $1,000 - 600 = 400$.

Say: “First choose the lower-cost outcome. Then ask who pays.”

Step 2: farmer has the right to undamaged crops

The rancher must prevent or compensate the damage. Compare paying 1,000 in compensation with building the 600 fence. The rancher chooses the fence.

Say: “The rancher bears the cost of the damage, so avoiding the damage is privately worthwhile.”

Do not add compensation to crop damage when calculating social cost. The damaged crop is a resource loss; compensation is a transfer between parties.

Step 3: rancher has the right to let cattle roam

The farmer otherwise bears 1,000 of damage. If permitted to install the fence directly, the farmer is willing to spend 600 to prevent it. Alternatively, the farmer can pay the rancher to install it.

Write “Illustrative payment: 800.” Have students calculate gains relative to no fence:

$$\text{Farmer's gain} = 1,000 - 800 = 200$$

$$\text{Rancher's gain} = 800 - 600 = 200.$$

Payments strictly between 600 and 1,000 make both parties better off in this bargain relative to no agreement. Endpoints leave one party indifferent. The exact division depends on bargaining; 800 is not a predicted unique payment.

Ask: “What stayed the same? What changed?” **Answer:** the fence is built in both rights cases; who pays and the division of gains change.

Step 4: change one number

Replace the fence cost with 1,400. Give students 30 seconds to choose the efficient outcome.

Answer: accept 1,000 of damage, assuming there is no cheaper remedy. Under farmer protection, the rancher compensates the farmer. Under the rancher’s right to roam, the farmer bears the loss. Neither party can create a gain by spending 1,400 to prevent 1,000 of harm.

Say: “Bargaining can lead to an efficient outcome. That outcome may include damage.”

Instructor cue: slide 19’s statement that someone builds a fence needs the condition that the fence is worth its cost.

27–33 minutes | When bargaining is difficult — slide 20

State the theorem in simple language: “If rights are clear and enforceable, and bargaining is costless, parties can reach an efficient agreement.” For this example, assume the parties can make the relevant payments and the assigned rights do not change the underlying costs or benefits.

Define **transaction costs**: finding the parties, negotiating, monitoring, and enforcing the agreement.

Return to the 600 fence and 1,000 damage. Add 500 in unavoidable negotiation and enforcement costs to the negotiated fence arrangement:

$$\text{Gain before transaction costs} = 400, \quad \text{gain after these costs} = 400 - 500 = -100.$$

Say: “This negotiated arrangement now costs more than it saves.” This calculation assumes those extra costs are necessary for that arrangement; it does not rule out a cheaper direct solution.

Ask: “Why is a bargain harder with ten thousand affected residents?” Expected answers: identifying everyone, disagreement over harm and payment, holding out, relying on others to pay, and enforcement.

Bridge: “Negotiation depends on who can control access. We can use access and use to classify goods.”

33–43 minutes | Board example 2: four types of goods — slides 21–27

Define the two terms one at a time:

Excludable: “It is feasible to prevent someone from using the good.” Payment gates, tickets, and passwords are examples.

Rival: “One person’s use leaves less quantity or quality for other people.” Eating a particular sandwich is rival. Watching a broadcast does not use up the signal.

Draw an empty matrix. Ask both questions before writing a category name.

	Rival	Nonrival
Excludable	Private good: sandwich	Club good: uncrowded subscription broadcast
Nonexcludable	Common resource: open-access fishery	Public good: national defense

Work aloud through the fishery: “Can users be kept out under open access? No. Does my catch leave fewer fish for others? Yes. It is a common resource.”

Work aloud through the broadcast: “Can a provider require a subscription? Yes. Does another viewer consume the signal? No. It is a club good.”

Use slide 24. The intended answer is the free outdoor light display, assuming viewers cannot practically be excluded and additional viewers do not crowd each other out.

Ask: “Is a government-provided sandwich a public good?” **Answer:** no. The sandwich is still rival and excludable. Who provides it does not determine its category.

Ask: “Can a private good have a positive externality?” **Answer:** yes. An individual vaccination dose is rival and excludable, while reduced transmission can benefit others. Classification and spillovers are different questions.

43–50 minutes | Board example 3: one road, four cases — slides 26–28

Give pairs two minutes to fill the final column, then discuss. State that an enforced toll makes exclusion feasible and that an unrestricted road in this exercise has no effective access restriction.

Road conditions	Reasoning	Classification
No toll, little traffic	No effective exclusion; another car creates negligible delay	Approximately public good
Toll, little traffic	Exclusion feasible; another car creates negligible delay	Approximately club good
No toll, congested	No effective exclusion; another car delays others	Common resource
Toll, congested	Exclusion feasible; another car delays others	Private-good characteristics

Say: “The physical road is the same. Access rules and congestion change its economic characteristics.”

Use slide 28. Expected textbook answer: club good. Add: “That describes access to an uncrowded gym. A machine with a queue is rival.” Accept a qualified answer supported by both tests.

Preview free riding: “If people cannot be excluded, they may wait for others to pay.” Save the numerical funding example for Class 3.

50–55 minutes | Exit check

Allow three minutes to write and two minutes to review.

1. A neighbor’s equipment causes 500 in annual harm. Soundproofing costs 200 annually and eliminates the harm. With costless bargaining, what is efficient? If the neighbor has the right to make noise, give one payment that benefits both parties relative to no agreement.
2. Classify an open-access fishery using both criteria.
3. Does assigning different rights necessarily change the efficient physical outcome in our bargaining model?

Answers: soundproofing; a payment of 350 from the affected person to the neighbor gives each a gain of 150. Fishery: nonexcludable and rival, hence common resource. Under the stated assumptions, changing rights changes distribution but not the efficient physical outcome.

Closing sentence: “Next time we will study why shared benefits can be underfunded and shared resources can be overused.”

Pacing and optional depth

If behind: reduce the bicycle discussion to one minute and demonstrate two road cases instead of four. Keep the expensive-fence variation: it prevents a central misconception about bargaining.

Optional extension: add 400 in unavoidable transaction costs to the exit-check soundproofing bargain. The gross gain is 300, so that arrangement no longer creates a net gain. Ask whether a different institution could achieve the outcome more cheaply; students need not assume one always exists.

Misconceptions to listen for: efficient means fair; property rights guarantee a fence; compensation removes the physical harm; government provision makes something a public good; nonrival means unlimited capacity in all circumstances.

Chapter 7 — Class 3

Public goods, common resources, and pollution policy

Instructor preparation

Length: 55 minutes. **Slides:** revisit 23, then 29–36. **Reading:** Chapter 7 notes, “Public Goods and the Free-Rider Problem” through the chapter review.

Learning goals: students can explain underprovision and overuse, evaluate a shared project, show how permit trading lowers cleanup cost, and distinguish the amount of cleanup from its allocation across firms.

Board setup: use three sections for the shared-project calculation, the two-firm permit table, and a cleanup graph. Write “Payments between parties are transfers” above the permit table.

0–5 minutes | Retrieve the two goods problems

Ask students to classify a widely visible light display and an open-access fishery. Require both criteria. Write:

Good	What another user’s participation does	Likely incentive problem
Public good	Receives a shared benefit without using it up	Wait for others to pay
Common resource	Takes some of a scarce shared resource	Use more before someone else does

Say: “Both are hard to exclude people from. Rivalry explains why one tends to be underprovided and the other overused.”

5–14 minutes | Board example 1: fund a shared display? — slides 23 and 29

Assume a display can be seen by three households, nonpayers cannot be excluded, and viewing is nonrival. Values measure each household’s total willingness to pay for this one display. There are no other benefits or harms in this illustration.

Household	Value of one display
A	40
B	35
C	25
Total	100

$$\text{Cost} = 80, \quad \text{net social benefit} = 100 - 80 = 20.$$

Ask: “Is the project worth doing?” **Answer:** yes, under these assumptions.

Ask: “Will one household pay the whole cost?” **Answer:** no household values it at 80. Joint funding can work, but voluntary funding is not assured.

Say: “A person can want the display and still prefer that other people pay. Low contributions do not necessarily mean low total benefit.”

Show a feasible cost split: A pays 30, B pays 30, C pays 20. Contributions sum to 80, and individual net gains are 10, 5, and 5. This proves mutually beneficial funding is possible; it does not prove those voluntary payments will occur.

Ask: “What if each household must pay an equal share?” Each pays about 26.67; C’s value is only 25. An efficient project need not benefit every person under every funding arrangement.

End with measurement: “The values are known in our example. In practice, people may misstate them, and we must count opportunity costs and harms such as noise.” Tax funding can solve a collection problem without automatically solving the valuation problem.

14–21 minutes | How much of the public good?

Say: “Choosing whether to provide one project is a total-benefit question. Choosing how much to provide requires marginal benefits and marginal costs.”

For one additional segment of a shared display, suppose household marginal values are 20, 15, and 10. Write:

$$\text{Community marginal benefit} = 20 + 15 + 10 = 45.$$

Everyone receives the same additional segment, so add dollar values for that unit. This is **vertical addition**. In a private-good demand calculation, by contrast, we add people’s quantities at the same price.

Now use the notes’ separate, simplified continuous model:

$$MB = 100 - 10Q, \quad MC = 40.$$

The curve is already the community total; do not add it three times. The numbers 20, 15, and 10 above only illustrate how to combine values for a single unit.

Ask students to solve:

$$100 - 10Q = 40 \Rightarrow Q^* = 6.$$

Draw: horizontal axis “Display quantity, Q ”; vertical axis “Dollars per additional unit.” Draw MB from $(0, 100)$ to $(10, 0)$ and horizontal $MC = 40$. Mark their intersection at $(6, 40)$.

Say: “Before six, more provision adds benefits greater than costs. Beyond six, more provision adds costs greater than benefits.” Treat quantity as divisible for this graph. With indivisible segments, compare each segment’s benefit and cost directly.

21–30 minutes | Board example 2: the common pasture — slides 30–34

Tell the deck’s story briefly: a pasture can support about 100 cows; 100 farmers each have one cow. Each farmer considers adding one. If all do so, there are 200 cows.

Say: “One farmer receives the gain from an extra cow. Some costs fall on everyone else through reduced grass and future productivity.”

Add this illustrative marginal calculation. These numbers are a separate teaching example, not estimates from the slides:

Extra cow's benefit to its owner = 50

Owner's full private marginal cost = 20

Additional damage to other users = 40.

Then calculate:

Private net gain = $50 - 20 = 30$

Social net gain = $50 - 20 - 40 = -10$.

Ask: "Why add the cow privately? Why avoid it socially?" **Answer:** the owner gains 30, but the combined effect is a loss of 10. The owner does not bear all costs.

Take two proposals to change incentives. Possible answers:

- An enforceable limit on cows or grazing days.
- A use charge reflecting marginal harm.
- Defined use rights with monitoring.
- Community rules with agreed sanctions for violations.

For each proposal ask: "Who checks whether people follow the rule?" Mention the notes' discussion of Ostrom: communities can sometimes manage shared resources through agreed rules and enforcement. Shared use does not imply inevitable collapse.

Bridge: "Pollution also uses a shared resource: the environment's ability to absorb emissions. A permit system limits that use."

30–44 minutes | Board example 3: trading pollution permits — slide 35

Budget about 3 minutes for setup, 3 for equal reductions, 5 for trading and payments, and 3 for interpretation.

Assume both firms initially emit 100 units. Combined emissions are 200. The cap allows 100, so total required cleanup is 100. Firm A can clean up any of its first 100 units at 10 per unit; B can do so at 2 per unit. Damage per emitted unit is the same across firms, trading is costless, and the cap is enforced.

Write first:

Emissions = initial emissions – cleanup.

Permits authorize emissions, not cleanup. This distinction prevents most errors in the exercise.

Firm	Initial emissions	Cleanup	Final emissions	Cleanup cost
------	-------------------	---------	-----------------	--------------

Step 1: equal reductions

Firm	Initial emissions	Cleanup	Final emissions	Cleanup cost
A	100	50	50	500
B	100	50	50	100
Total	200	100	100	600

Ask: “Can we achieve the same total emissions with a lower resource cost?” Let students identify B as the cheaper cleaner.

Step 2: allow trading

Give each firm 50 emissions permits. Let B clean up all 100 units and sell its 50 permits to A.

Firm	Permits after trade	Cleanup	Final emissions	Cleanup cost
A	100	0	100	0
B	0	100	0	200
Total	100	100	100	200

$$\text{Resource saving} = 600 - 200 = 400.$$

Say: “Total emissions are still one hundred. The low-cost firm does more cleanup. That reduces the total cost of meeting the cap.”

Step 3: show why both firms accept a trade

Use an illustrative price of 6 per permit. A buys 50 permits for 300; B receives 300.

Firm	Cost under equal reductions	Cleanup cost after trade	Permit payment, paid (+) or received (–)	Net cost after trade	Gain from trade
A	500	0	+300	300	200
B	100	200	–300	–100	200

Explain B’s negative net cost: its permit receipts exceed its cleanup expense. This is possible because B initially received valuable permits for free.

Any price strictly between 2 and 10 supports gains for both firms from this trade. The example does not determine a unique price of 6.

Ask: “Why are the 300 dollars of permit payments not an additional social cost?” **Answer:** A pays what B receives. The resource cost is the 200 spent on cleanup.

Ask: “Is a free permit costless to use?” **Answer:** no. Using it sacrifices the revenue from selling it.

Say: “Trading lowers the cost of meeting this cap. It does not tell us whether one hundred units is the best cap.”

44–50 minutes | Return to the opening pollution question — slide 36

Use a separate illustrative cleanup model. Do not combine its units or numbers with the two-firm example.

Let initial pollution be 10 units and let A denote units of pollution eliminated, with $0 \leq A \leq 10$:

$$MB_{\text{cleanup}} = 12 - A, \quad MC_{\text{cleanup}} = 2 + A.$$

Solve with the class:

$$12 - A = 2 + A \Rightarrow A^* = 5.$$

Remaining pollution is $10 - 5 = 5$.

Draw: horizontal axis “Cleanup, A ”; vertical axis “Dollars per cleanup unit.” Draw downward MB through $(0, 12)$ and upward MC through $(0, 2)$. Mark the intersection $(5, 7)$ and full cleanup at $A = 10$.

Say: “Moving right means more cleanup and less pollution. Before five, cleanup prevents more damage than it costs. Beyond five, extra cleanup costs more than the damage it prevents.”

At full cleanup, marginal benefit is 2 and marginal cost is 12 in this example. This is why zero pollution is not the efficient target here.

Ask: “Could full cleanup ever be efficient?” **Answer:** yes, if its marginal benefits are at least as large as marginal costs throughout the feasible cleanup range. The model’s values determine the answer; zero is not ruled out in advance.

50–55 minutes | Exit check

Allow three minutes individually, then two minutes to review.

1. Three households value one shared warning system at 60, 50, and 40. It costs 120. Find its net social benefit and explain why voluntary funding might fail.
2. Explain in one sentence why permit trading can lower cost without raising total emissions.
3. If one more unit of cleanup prevents 3 of damage but costs 8, should it be undertaken on these numbers alone?

Answers: benefit 150, net benefit 30; households may try to benefit without contributing. Trading shifts cleanup toward lower-cost firms while the enforced cap fixes total permitted emissions. Do not undertake the extra cleanup unit: its marginal cost exceeds its marginal benefit by 5.

Closing sentence: “For each policy, ask which cost or benefit is missing from private decisions, and whether the proposed rule changes that incentive.”

Pacing and optional depth

If behind: omit the equal-share display discussion and the public-good graph, but keep the vertical-addition explanation. In permit trading, calculate costs aloud and omit the final net-cost table if needed. Protect the final cleanup comparison and exit check.

Optional independent practice: two firms each emit 80 units; the cap is 80; constant cleanup costs are 9 for A and 3 for B. Equal reductions cost $40(9) + 40(3) = 480$. Having B clean up all 80 costs 240. With 40 permits initially assigned to each, B sells 40 to A. At an illustrative price of 6, A pays 240 instead of spending 360 on cleanup; B spends 240 and receives 240 instead of spending 120. Each gains 120. Total emissions remain 80.

Optional discussion: with rising marginal cleanup costs, lowest-cost allocation generally equalizes firms' marginal cleanup costs for interior choices. With constant costs in today's example, concentrate cleanup at B until its capacity is exhausted. Location-dependent pollution harm can make emissions at the two plants environmentally different even if the total is unchanged.

Misconceptions to listen for: public goods have no cost; a valuable project automatically receives voluntary funding; common resources are nonrival; a permit authorizes cleanup; trading changes the cap; zero pollution is always or never efficient.